Cheatsheet AZ-104 — 01 · Microsoft Entra ID: users, groups, licensing, guests and SSPR

1. Tenant, editions and domains

Feature	Free	P1	P2
Users, groups, B2B, Security Defaults, device registration, basic SSO	✓	✓	✓
Smart Lockout and global banned password list	✓	✓	✓
SSPR: password **change** (user knows the password)	✓	✓	✓
SSPR: password **reset** for cloud users	✗ (yes with M365 Business Standard or higher)	✓	✓
SSPR with **writeback** to AD DS	✗	✓	✓
Dynamic groups (users or devices)	✗	✓	✓
Group-based licensing	✗ (yes with Office 365 / M365 E3 or higher)	✓	✓
Role-assignable groups (Entra roles to groups)	✗	✓	✓
Administrative units (license for each delegated admin)	✗	✓	✓
Conditional Access, Application Proxy, Entra Connect Health	✗	✓	✓
Self-service group management, **expiration** and **naming policy** for M365 groups	✗	✓	✓
Password Protection: **custom** banned list and on-prem agent	✗	✓	✓
PIM, **Identity Protection**, **Access Reviews**	✗	✗	✓

Rule: P1 is included in Microsoft 365 E3 / EMS E3; P2 in Microsoft 365 E5 / EMS E5. If the wording says "minimize licensing costs" and asks for dynamic groups, CA or writeback, the minimum answer is **P1**; if it asks for PIM or risk detection, **P2**.

- **Custom domain:** verified with a **TXT** (or MX) record in public DNS; the initial `<tenant>.onmicrosoft.com` cannot be deleted or renamed. A domain can be verified in **one** tenant only; subdomains of a verified domain are verified automatically. "Make primary" sets the default UPN suffix.
- **User settings:** "Restrict non-admin users from creating tenants" (recommended Yes), "Users can register applications", "Restrict access to Microsoft Entra administration portal". A user can be a **member** of their home tenant and a **guest** in others.
- **Global Administrator ≠ subscription Owner.** A GA does not see Azure resources by default: it needs **elevate access** (User Access Administrator at root scope; see cheatsheet 02). Microsoft recommends fewer than 5 GAs.

2. Users: types, lifecycle and roles

	Cloud-only	Synchronized (hybrid)	Guest (B2B)
Created	Portal, bulk CSV, PowerShell / CLI	In AD DS; brought in by Entra Connect / Cloud Sync	Invitation (portal, CSV, `New-MgInvitation`)
Edited	In Entra ID	**Only in AD DS** (attributes greyed out in the portal); cloud exceptions: licenses, roles, usage location	In Entra ID (local attributes) and in their home tenant
userType	Member	Member	**Guest** (UPN `name_domain.com#EXT#@tenant.onmicrosoft.com`)
Authenticates in	Entra ID	Entra ID (PHS) / on-prem (PTA, federation)	Their provider: Entra, MSA, Google, SAML/WS-Fed, email code
Signal	"cloud identity", "no on-premises"	"synchronized from AD DS", "attribute is read-only"	"partner", "external user", "collaborate"

- **Bulk operations** (Users → Bulk operations): CSV template for **create**, **invite**, **delete** and **download**. The template columns cannot be changed.
- **Deletion:** the user stays in **Deleted users for 30 days** and can be **restored** with properties and memberships; afterwards (or with "Delete permanently") it is gone. "Block sign in" when the requirement is to block access without deleting.
- **Usage location** is mandatory to assign a license and **is not synchronized** from AD DS → "license cannot be assigned" error on a synced user.
- Admin reset → temporary password + "change password at next sign-in". If the wording asks for the "least privileged role", use the table:

Role	Can reset the password of	Also
Password Administrator	Users with no role, Directory Readers, Guest Inviter, Password Admin	Nothing else: it is the **minimum** for "reset passwords of users"
Helpdesk Administrator	The above + Helpdesk Admin, Message Center Reader, Reports Reader	Invalidates refresh tokens, support tickets, reads BitLocker keys
Authentication Administrator	Users with no role, readers, Password Admin, Auth Admin	Manages **MFA methods** of non-admins (re-register, delete methods)
User Administrator	Non-admins + Groups Admin, Helpdesk Admin, Password Admin, User Admin, readers	**Creates and manages users and groups**, assigns **licenses**, restores deleted users
Privileged Authentication Administrator	**Anyone**, including Global Admins and members of role-assignable groups	MFA methods of any user

Role	Can reset the password of	Also
Global Administrator	Anyone	Everything; can also elevate access over Azure

Other roles that show up: **Guest Inviter** (invite only), **License Administrator** (licenses and usage location only), **Groups Administrator** (create and manage groups and their settings), **Privileged Role Administrator** (assigns Entra roles, creates role-assignable groups and administrative units), **Global Reader** (read everything), **Cloud Device Administrator** (devices).

3. Groups

	Security	Microsoft 365
Members	Users, **groups**, **devices**, service principals	**Users only** (no groups, no devices)
Purpose	Resource access, Azure RBAC, licensing, CA, SSPR	Collaboration: mailbox, SharePoint, Teams, Planner
Dynamic	Users **or** devices (never mixed)	Users only
Role-assignable	Yes (**assigned** membership)	Yes (**assigned** membership)
Expiration / naming policy	No	Yes (P1)
Restore after deletion	Cloud groups: 30 days (preview); Exchange ones (distribution, mail-enabled security) **cannot be restored**	**30 days** (Deleted groups)
Azure RBAC	Yes (transitive nesting)	Yes

Membership type	License	Rules
Assigned	None	Manual or by owners. The only valid type for **role-assignable** groups
Dynamic user	**P1 for each unique user** who is a member of any dynamic group	Rule on `user.*`; members cannot be added manually; rule builder up to 5 expressions; text up to 3,072 characters
Dynamic device	None (devices do not consume P1)	Rule on `device.*`, only in security groups and only via the text box

```
user.department -eq "Sales"
(user.department -eq "Sales") -and (user.country -eq "ES")
user.userType -eq "Guest"                                  → all guests
(user.objectId -ne null) -and (user.userType -eq "Member")   → all members
user.department -in ["Sales","Marketing"]
user.displayName -match "^Da.*"
(user.proxyAddresses -any (_ -startsWith "SMTP:sales"))
(device.deviceOSType -eq "Windows") -and (device.deviceOSVersion -startsWith "10.0.1")
device.deviceOwnership -eq "Company"
device.deviceTrustType -eq "ServerAD"                      → hybrid joined
Direct Reports for "<manager objectId>"                    → cannot be combined with other rules
```

Operators: `-eq -ne -startsWith -notStartsWith -endsWith -notEndsWith -contains -notContains -match -notMatch -in -notIn`, logical `-and -or -not`, collections `-any -all`. Maximum **15,000 dynamic groups** per tenant. Switching assigned ↔ dynamic is possible and recalculates the membership.

- **Role-assignable** ("Microsoft Entra roles can be assigned to the group" = Yes): only **at creation**, immutable, requires **P1**, **assigned** membership, maximum **500** per tenant, no nesting, created by a **Privileged Role Administrator** and its members can only be changed by PRA / GA or owners. For a **temporary** role on the group → PIM (P2).
- **Nesting:** Azure RBAC and Conditional Access evaluate nested groups; **group-based licensing does not** (first level only); Entra roles **do not**; dynamic groups **do not contain groups** (except `memberOf`, preview); M365 groups cannot be nested.
- **Group settings** (Groups → General): "Users can create security groups / Microsoft 365 groups in Azure portals", "Owners can manage group membership requests in My Groups" (self-service, P1), **expiration** of M365 groups (180 / 365 / custom ≥ 30 days; owners renew by email; auto-renewed with activity in Teams, SharePoint or Outlook) and **naming policy** (prefix / suffix + blocked words).

4. Licensing

	Direct assignment	Group-based
Requirement	Any edition	**P1/P2** or an Office 365 / Microsoft 365 E3 or higher plan
Usage location	**Mandatory** (error if missing)	If the user has none, inherits the **tenant** location
Nesting	—	**No**: only first-level members receive the license
Removing it	From the user	From the **group** or by removing the user; an **inherited license cannot be removed on the user**
When leaving the group	Kept	Removed, unless the user also has it **directly** (then only the inheritance is lost)
Errors	Not enough licenses, Usage location not allowed	Same + **Conflicting service plans**, **Other products depend on this license**, **Proxy address conflict** → fix and **Reprocess**
Who	License Administrator, User Administrator, Global Admin	Same + Groups Administrator

- **Service plans can be disabled** (e.g. Yammer) on the group assignment: "Turn apps and services on or off".

- Move a user between licensed groups **without interruption**: add to the destination → confirm the license → remove from the source.
- PowerShell: `Update-MgUser -UserId u -UsageLocation ES` → `Set-MgUserLicense -UserId u -AddLicenses @{SkuId="..."} -RemoveLicenses @()`; for groups `Set-MgGroupLicense`; available SKUs with `Get-MgSubscribedSku`.

5. External users (B2B)

Setting (External Identities → External collaboration settings)	Options	Signal
Guest user access	Same as members · **Limited** (default: cannot enumerate users or groups) · **Restricted** (own profile only)	"guests must not see other users" → Restricted
Guest invite settings	Anyone incl. guests (default) · Members + admin roles · **Only specific admin roles** (User Administrator, **Guest Inviter**) · No one	"only the help desk can invite" → Only admin roles + Guest Inviter
Collaboration restrictions	Any domain · **Deny** list · **Allow** list	"only invite users from fabrikam.com" → Allow list
External user leave settings	Yes / No (requires the tenant's privacy information)	"guests can remove themselves"
Self-service sign-up via user flows	Yes / No	"users sign up to our app and get a guest account"
Cross-tenant access settings	Inbound / outbound per tenant, users, groups and apps; **trust** MFA, compliant and hybrid joined from the partner tenant	"guests prompted twice for MFA" → trust the partner's MFA
Identity providers	Entra ID, Microsoft account, Google, Facebook (self-service only), **SAML/WS-Fed** (domains without Entra), **email one-time passcode** (fallback)	"partner has no Microsoft account" → email OTP

- Flow: invite → status **Pending acceptance** → the guest **redeems** with their provider → **Accepted**. If they do not redeem: "Resend invitation"; if they change identity: "Reset redemption status".
- A guest is a normal object: it joins groups (also dynamic ones with `user.userType -eq "Guest"`), receives Azure RBAC and Entra roles, and Conditional Access targets it with "All guest and external users". Convert guest → member: change **User type** (no need to re-invite).
- External ID billing by **MAU**: the first **50,000** monthly active users are free.
- **B2B direct connect** (Teams shared channels) creates no guest object. **External ID for customers (formerly B2C)** is for consumer apps, not for business-to-business collaboration. **Lighthouse** manages resources in other tenants without guest accounts.

6. SSPR and password protection

Blade	Setting	Exam detail
Properties	None · **Selected** · All	Selected accepts **a single group** in the portal; **nested** groups count. Typical use: pilot
Authentication methods	Number of methods: **1 or 2** · App (notification / code), email, mobile phone (SMS / call), office phone, **security questions** (3-5 to register, 3-5 to answer)	The app counts as **one** method even if you enable notification and code. Questions **not available to admins**. Since Sept. 2025 methods are managed in the **Authentication methods policy** (migration from the legacy policy)
Registration	Require users to register when signing in (Yes / No) · Days before re-confirm: **0-730**, default **180** (0 = never)	"users must provide contact info at next sign-in" → Yes
Notifications	Notify users on password resets (default **Yes**) · Notify **all admins** when other admins reset (default **No**)	"all admins alerted when an admin resets" → second toggle
Customization	Custom helpdesk link (email or URL)	"Contact your administrator"
On-premises integration	**Write back passwords** to on-prem · Allow users to **unlock without resetting**	Requires Entra Connect (or Cloud Sync) with password writeback enabled and permissions in AD DS

- **Administrators:** always enabled for SSPR, with the **two-gate** policy (2 methods) and no security questions, even if Properties is None or Selected.
- **Licensing:** password change → anyone; cloud **reset** → M365 Business Standard or higher, P1/P2; **writeback** → **P1/P2 or M365 Business Premium**. Without writeback, a synced user cannot reset from the cloud.
- **Combined registration** (`aka.ms/mysecurityinfo`): MFA and SSPR are registered together. **Smart Lockout** (Free): locks after **10** failed attempts for **60 s** by default. **Password Protection**: global banned list in Free; custom list and agent for AD DS with **P1**.

7. Hybrid identity, devices and administrative units

	Entra Connect Sync	Entra Cloud Sync
Where it is configured	On-prem server (single active server + staging)	**In the cloud**; lightweight agents, **several active** (automatic HA)
Forests	Connected	Connected and **disconnected** (mergers)
PHS · password writeback · device sync (hybrid join) · Exchange hybrid	✓	✓
PTA and **ADFS** configuration	✓	✗ (configured separately)
Device writeback, advanced sync rules, full attribute filtering, cross-forest references	✓	✗ (OU-based / basic filtering only)
Group writeback v1	✓	✓

	Entra Connect Sync	Entra Cloud Sync
Group provisioning **to** AD DS, on-demand provisioning	✗	✓
Limits	Unlimited; groups up to 250,000 members	**150,000 objects per domain**; groups up to 50,000 members

- Synced objects are edited **in AD DS**; delta cycle every **30 min** (`Start-ADSyncSyncCycle -PolicyType Delta`, `Initial` for a full one); password hashes travel every **2 min**. Non-routable UPN suffixes (`.local`) → add a public suffix before syncing. **Entra Connect Health** (P1) alerts on sync failures.

	Entra registered	Entra joined	Hybrid joined
Device	**Personal / BYOD**	Corporate, **cloud-only**	Corporate, **joined to AD DS**
OS	Windows 10+, iOS, Android, macOS	Windows 10/11, Windows Server 2019+ (VMs), macOS, Linux	Domain-joined Windows
Sign-in	Local account or MSA + "Add work account"	**Entra ID** account	**AD DS** account
How	Settings → Access work or school; MDM enrollment	OOBE, Settings, Autopilot, bulk enrollment	GPO / SCP + Entra Connect or Cloud Sync (device sync)
trustType	Workplace	AzureAD	ServerAD

- **Device settings:** "Users may join devices" (default **All**; does not affect hybrid join or Azure VMs), "Users may register their devices" (All / None; MDM forces All), "Require MFA to register or join" (default **No**; prefer the CA user action), **Maximum number of devices per user** (default **50**, up to 100 or Unlimited; does not apply to hybrid), local admins (**Microsoft Entra Joined Device Local Administrator** role), **Enterprise State Roaming** (P1).
- **Disable** revokes the PRT and blocks access; **Delete** is irreversible and removes the **BitLocker keys**. Done by **Cloud Device Administrator** (or Intune Administrator). BitLocker keys: the device owner or Cloud Device / Helpdesk / Intune / Security Admin / Security Reader.
- **Administrative units (AU):** container of **users, groups and devices** to delegate an Entra role over them only (e.g. Helpdesk Administrator for the Bilbao office). Cannot be nested; assigned or **dynamic** membership (P1 per member); **P1 for each admin** holding a role scoped to the AU; adding a **group** to the AU does not bring in its **members**. **Restricted management AU**: its members can only be managed by admins holding a role scoped to the AU, not by tenant-level admins.

8. Commands that show up

Task	PowerShell (Microsoft Graph / Az)	Azure CLI
Create user	`New-MgUser -DisplayName -UserPrincipalName -MailNickname -AccountEnabled -PasswordProfile @{Password="..."; ForceChangePasswordNextSignIn=$true}` · `New-AzADUser`	`az ad user create --display-name --user-principal-name --password`
Create group	`New-MgGroup -DisplayName -MailNickname -SecurityEnabled -MailEnabled:$false` · `New-AzADGroup`	`az ad group create --display-name --mail-nickname`
Add member	`New-MgGroupMember -GroupId -DirectoryObjectId` · `Add-AzADGroupMember`	`az ad group member add --group --member-id`
Invite guest	`New-MgInvitation -InvitedUserEmailAddress -InviteRedirectUrl`	`az rest --method post --url https://graph.microsoft.com/v1.0/invitations --body '{...}'`
License	`Update-MgUser -UsageLocation ES` → `Set-MgUserLicense -AddLicenses @{SkuId=...} -RemoveLicenses @()` · `Get-MgSubscribedSku`	No native command (via `az rest`)
Bulk create	`Import-Csv .\users.csv` piped into `ForEach-Object { New-MgUser ... }`	Portal → Bulk operations (CSV)
Restore deleted	`Restore-MgDirectoryDeletedItem -DirectoryObjectId`	Portal → Deleted users / Deleted groups
Sync now	`Start-ADSyncSyncCycle -PolicyType Delta` (on the Connect server)	—

The **AzureAD** and **MSOnline** modules (`New-AzureADUser`, `New-MsolUser`) are retired: the modern answer is **Microsoft Graph PowerShell**, although older questions still list them as a valid option.

9. Decision tree

```
What does the wording ask for?
├─ A user from another company accesses our resources ..... B2B invitation (guest)
├─ Invitations only to certain domains ..... Collaboration restrictions (allow list)
├─ Guests must not see the directory ..... Guest user access: Restricted
├─ Automatic membership by attribute ..... Dynamic group (P1)
├─ Automatic license when joining a group ..... Group-based licensing (+ usage location)
├─ Entra ID role to a group ..... Role-assignable group (at creation, assigned)
├─ Helpdesk for one office / department only ..... Administrative unit + role at that scope
├─ Reset passwords of regular users (least privilege) ..... Password Administrator
├─ Create users, groups and assign licenses ..... User Administrator
├─ SSPR for a pilot ..... Properties: Selected + one group
├─ SSPR that also changes the password in AD DS ..... Password writeback (P1 + Connect / Cloud Sync)
├─ Change the department of a synced user ..... In AD DS, then wait for or force the delta sync
└─ Device: personal / corporate cloud-only / joined to AD DS ..... Entra registered / Entra joined / Hybrid joined
```

10. Signals → answer

Wording	Answer
"minimize administrative effort", "users move between departments"	Dynamic group with `user.department`
"automatically assign licenses to members of the group"	Group-based licensing
"license cannot be assigned", "usage location"	Fill in **Usage location** (not synced from AD DS)
"remove license from user" fails	It is inherited: remove it on the group or remove the user from the group
"assign the Entra role to the group" and the group already exists	Impossible: create a new **role-assignable** group
"partner needs access", "no new accounts in their tenant"	B2B invitation
"only users from contoso.com can be invited"	Collaboration restrictions → Allow list
"guests must not enumerate users"	Limited (the default) already blocks it; Restricted if they must not see anyone's profile or memberships either
"least privileged role to reset passwords"	Password Administrator (Helpdesk if sessions must be invalidated)
"least privileged role to create users and groups"	User Administrator
"least privileged role to invite guests"	Guest Inviter
"least privileged role to manage devices"	Cloud Device Administrator
"regional helpdesk", "only users in the Paris office"	Administrative unit (dynamic by `city`)
"SSPR for a pilot group"	Selected + that group (only one)
"administrators must use two methods for SSPR"	Already the default (two-gate); nothing to configure
"users prompted to re-register every 90 days"	Registration → days before re-confirm = 90
"deleted a user 20 days ago"	Restore from Deleted users (30 days)
"attribute is grayed out" in the portal	Synced user: edit in AD DS
"Global Administrator cannot see the subscription"	Elevate access → User Access Administrator at root (cheatsheet 02)
"device limit reached"	Device settings → Maximum number of devices per user (default 50)
"BitLocker recovery key"	Entra ID → Devices → device → Show Recovery Key
"dynamic group for Windows devices"	Security group, dynamic device, rule `device.deviceOSType -eq "Windows"` via text box

11. Mistakes to avoid

1. Assigning a license without **Usage location**, or expecting the attribute to come from AD DS.
2. Creating the group and trying to mark it role-assignable afterwards: only **at creation**, and never dynamic.
3. Nesting groups for licensing: group-based licensing **does not inherit** through nested groups.
4. Assuming a Global Administrator sees Azure subscriptions and resources by default.
5. Editing attributes of a synced user in the portal (they are greyed out): they are changed in AD DS.
6. Selecting **several groups** in SSPR Selected: it accepts only one (nested ones do count).
7. Granting **User Administrator** when Password Administrator is enough, or Global Admin when Guest Inviter is enough.
8. Choosing External ID for customers (B2C) or Lighthouse to collaborate with **users** from another company: that is **B2B**.
9. Expecting Cloud Sync to configure **PTA** or perform **device writeback**: that is Connect Sync.
10. Adding a group to an administrative unit believing its members fall within the scope.